A-A-54239
27 August 1990

COMMERCIAL ITEM DESCRIPTION

DRILL, WIRE-PASS, BONE

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers a wire-pass drill suitable for use with a bur guard and rotary handpieces in oral and small bone surgical procedures.

Salient characteristics:

Material. The drills shall be fabricated from Class 4 type 440A martensitic corrosion-resisting steel in accordance with ASTM Designation F899 for composition only.

Hardness. The drills shall be properly hardened and passivated. The drills shall have a Rockwell hardness of C45 to C55, after proper stress relieving when tested in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

Style, design, and dimensions. The drills shall be of one-piece construction. The proximal end of the drill shall be "J"-notched for secure locking into "J"-notched type rotary handpieces, (see notes). The blades of the drill shall be uniformly cut, shaped, and sharpened to cut at the tip as well as the sides of the head, and to provide automatic chip clearance. The design and dimensions of the drills shall be in accordance with Figure 1 as specified herein. The drill shall be suitable for use in a bur guard which accommodates 64 mm drills.

Runout. The total indicated runout of the drills shall not exceed 0.003 inch when measured as specified herein.

Corrosion resistance. The drills shall show no evidence of corrosion after being subjected to the copper sulfate test of ASTM Designation F1089, and the autoclavability test as specified herein.

Concentricity and straightness. The drills shall meet the concentricity and straightness requirements of Figure 1 as specified herein.

AMSC N/A FSC 6515

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Finish. The surface roughness of the shank shall be no more than the root-mean-square (RMS) average deviation of 16 micro-inches in the longitudinal direction of the shank, when determined as specified herein. The operative (distal) end shall have a smooth, fine-honed finish with sharp cutting edges.

Identification marking. Each drill shall be marked with the manufacturer's part number. The part number shall be electro-etched onto each drill. The marking shall not adversely affect or interfere with the function, or operation of the drill, or its ability to fit into its associated handpieces. All markings shall be legible, permanent, and distinguishable to the unaided eye.

Workmanship. The "J"-notch shall be cleanly cut. The drills shall be free from untrimmed edges, burrs, or other defects which detract from their appearance or may impair their serviceability.

Unit. Each (EA). One drill, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. The unit of product for examination purposes shall be one drill. Sampling for examination shall be conducted in accordance with MIL-STD-105 at inspection level II, except for dimensions which shall be conducted at inspection level S-2, all with an AQL of 2.5 (percent defective).

Defects

Drills not of one piece construction.
Total indicated runout exceeds 0.003 inch.
Dimensions not as specified.
Drills not free from chips, cracks, or pits; workmanship not as specified.
Finish not as specified.

Examination. The drills shall be examined for compliance with this commercial item description.

Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with the gages.

Runout. The drill shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360-degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Visual examination. Examination for possible drill defects not perceptible to the unaided eye shall be made using a 10x magnification.

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Sampling for tests. The unit of product for test purposes shall be one drill. Sampling for tests shall be conducted in accordance with MIL-STD-105 at inspection level S-1, with an AQL of 1.0 (percent defective).

Tests. Tests shall be conducted to determine compliance with this document's requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Chemical composition. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151. In case of dispute, method 111 shall govern.

Surface roughness. The surface roughness of the drills shall be determined by using a Brush Surfindicator or other equipment of equal accuracy. The RMS average shall be determined by squaring the height measurements of the peaks and valleys of the surface of the shank, obtaining the averages of the squared values, and then extracting the square root of this square root.

Autoclaving. The test drill shall be cleaned in accordance with the instructions supplied by the manufacturer. The drill shall then be autoclaved at $270^0 \pm 5^0$F and 27 $\pm$ 1 pounds per square inch pressure for a minimum of 20 minutes. After the drill is removed from the autoclave, the entire surface area of the drill shall be examined for evidence of corrosion.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pounds units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTES:

1. The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6515-01-194-0383	DRILL, WIRE-PASS BONE, Corrosion-Resisting Steel, 1.5 mm Head Diameter, 64 mm Long

2. The following items are used with the drills:

National Stock Number	Item Identification
6515-01-171-5236	HANDPIECE, SURGICAL, Small Bone, Straight

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6515-01-167-2620 GUARD, SURGICAL BUR, 28 mm Long

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

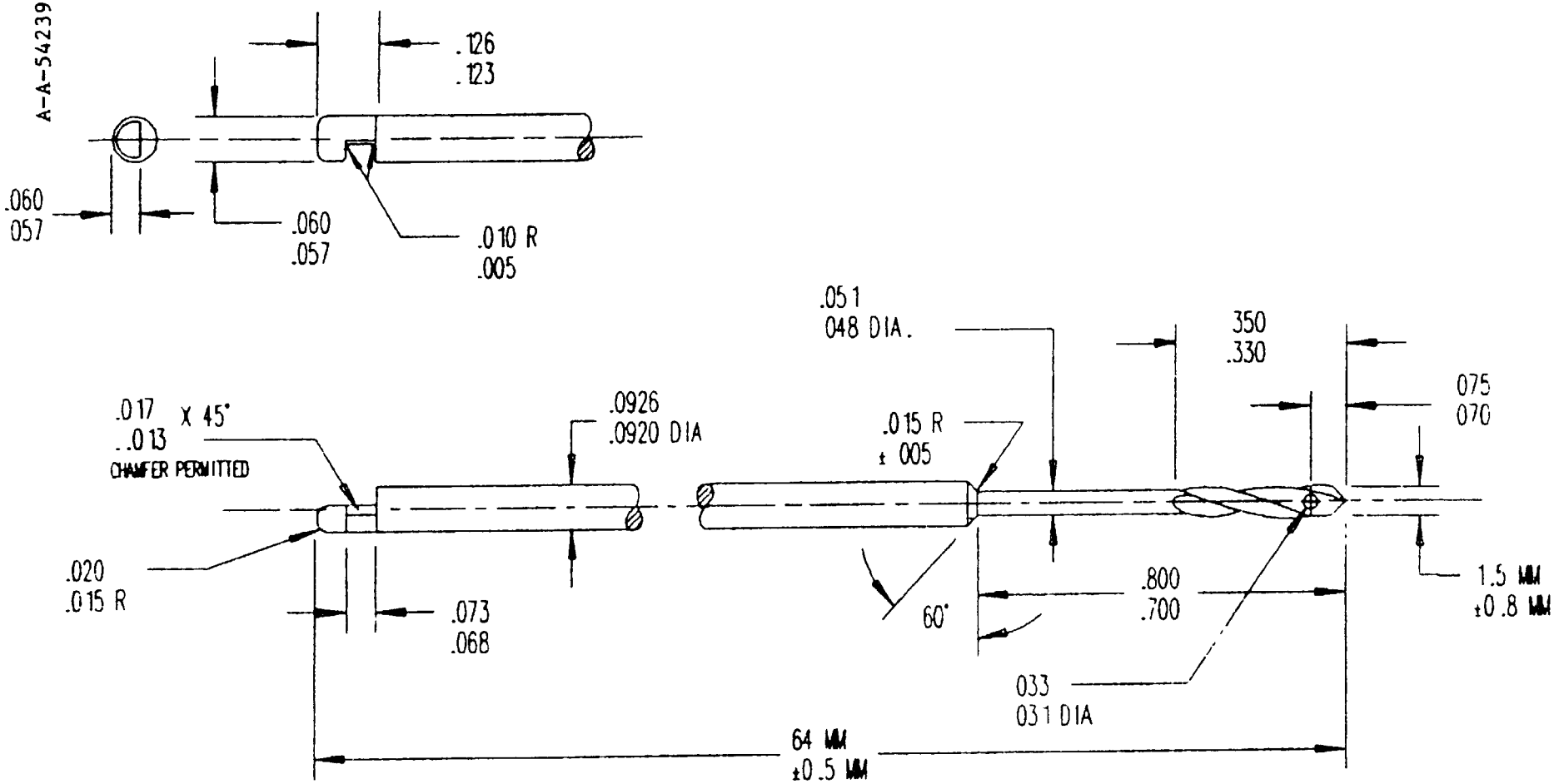
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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6515-4717
LOCATION: NCR DISK 2

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NOTES

1. SHANK FINISH NOT TO EXCEED 16 R.M.S.
2. BUR HD TO SHANK RUNOUT MUST BE LESS THAN .003 T.I.R.
3. BUR TO HAVE STD 2 FLUTES WHICH TURN 180 IN .340 LG.
4. BUR TO HAVE STD 118° END POINT.

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

FIGURE 1 DRILL,WIRE-PASS,BONE,CORROSION-RESISTING STEEL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-54239	DRILL, WIRE-PASS, BONE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street City State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* -- Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional
c MAILING ADDRESS *(Street City State ZIP Code)* -- Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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